ONLINE SALES REPORT

```
)
SET InvoiceDate = STR_TO_DATE(@InvoiceDate, '%m/%d/%Y %H:%i');

SELECT InvoiceNo, Description, Quantity
FROM retail_sales
LIMIT 10;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

InvoiceNo	Description	Quantity
536365	WHITE HANGING HEART T-LIGHT HOLDER	6
536365	WHITE METAL LANTERN	6
536365	CREAM CUPID HEARTS COAT HANGER	8
536365	KNITTED UNION FLAG HOT WATER BOTTLE	6
536365	RED WOOLLY HOTTIE WHITE HEART.	6
536365	SET 7 BABUSHKA NESTING BOXES	2

```
    SUM(InvoiceDate IS NULL) AS InvoiceDate_Missing,
    SUM(UnitPrice IS NULL) AS UnitPrice_Missing,
    SUM(CustomerID IS NULL) AS CustomerID_Missing,
    SUM(Country IS NULL) AS Country_Missing
FROM retail_sales;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

InvoiceNo_Missing	StockCode_Missing	Description_Missing	Quantity_Missing	InvoiceDate_Missing	UnitPrice_Missing	CustomerID_Missing	Cour
0	0	0	0	0	0	0	0

```
    SUM(UnitPrice IS NULL) AS UnitPrice_Missing,
    SUM(CustomerID IS NULL) AS CustomerID_Missing,
    SUM(Country IS NULL) AS Country_Missing
FROM retail_sales;

DELETE FROM retail_sales
WHERE CustomerID IS NULL;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

InvoiceNo_Missing	StockCode_Missing	Description_Missing	Quantity_Missing	InvoiceDate_Missing	UnitPrice_Missing	CustomerID_Missing	Cour
0	0	0	0	0	0	0	0

```sql
USE online_retail;
-- Negative quantities (returns, but must be handled carefully)
SELECT * FROM retail_sales WHERE Quantity < 0;

-- Zero or negative prices
SELECT * FROM retail_sales WHERE UnitPrice <= 0;
```

Result Grid

InvoiceNo	stock_code	Description	Quantity	InvoiceDate	unit_price	customer_id	Country
C536379	D	discount	-1	2010-12-01 09:41:00	27.50	14527	UNITED KINGDOM
C536383	35004C	set of 3 coloured flying ducks	-1	2010-12-01 09:49:00	4.65	15311	UNITED KINGDOM
C536391	22556	plasters in tin circus parade	-12	2010-12-01 10:24:00	1.65	17548	UNITED KINGDOM
C536391	21984	pack of 12 pink paisley tissues	-24	2010-12-01 10:24:00	0.29	17548	UNITED KINGDOM
C536391	21983	pack of 12 blue paisley tissues	-24	2010-12-01 10:24:00	0.29	17548	UNITED KINGDOM
C536391	21980	pack of 12 red retrospot tissues	-24	2010-12-01 10:24:00	0.29	17548	UNITED KINGDOM
C536391	21484	chick grey hot water bottle	-12	2010-12-01 10:24:00	3.45	17548	UNITED KINGDOM
C536391	22557	plasters in tin vintage paisley	-12	2010-12-01 10:24:00	1.65	17548	UNITED KINGDOM

Key points:

- Each row represents a **line item** in an invoice (one product in a transaction).
- Multiple rows can share the same `InvoiceNo` if several products were bought in one order.
- It mixes **numerical features** (Quantity, UnitPrice, CustomerID) with **categorical features** (InvoiceNo, StockCode, Description, Country).
- Useful for **sales analysis, customer segmentation, and market basket analysis**.

In short, it's a **real-world e-commerce dataset** that captures what was bought, when, by whom, and at what price.

Explanation of the chosen method

For the **Online Retail** dataset:

- **CustomerID:** Rows with missing `CustomerID` are often removed if the analysis focuses on customer behavior (e.g., customer segmentation or purchase frequency), because the customer cannot be identified.
- **Description:** Missing descriptions can be replaced with `"Unknown"` since they are descriptive text and do not affect numerical calculations.
- **UnitPrice:** If missing, replacing with the **average** is generally better than using `0`, because `0` may incorrectly suggest the item was free. However, if a missing value truly indicates no charge, `0` may be appropriate.

Recommended approach for this dataset:

- Replace missing `Description` values with `"Unknown"`.
- Remove rows with missing `CustomerID` if customer-level analysis is required.
- Replace missing numerical values (such as `Unit_Price`, if any) with the column average rather than `0`, unless there is a business reason to use zero.

Issue Found	Action Taken
Missing Values	Filled/Removed
Duplicates	Removed
Invalid Entries	Corrected
Standardization	Applied

Figure 1: Total Sales by Country (Bar Chart)

Explanation:
The chart compares total revenue generated by each country. The United Kingdom contributes the largest share of sales, indicating it is the company's primary market.

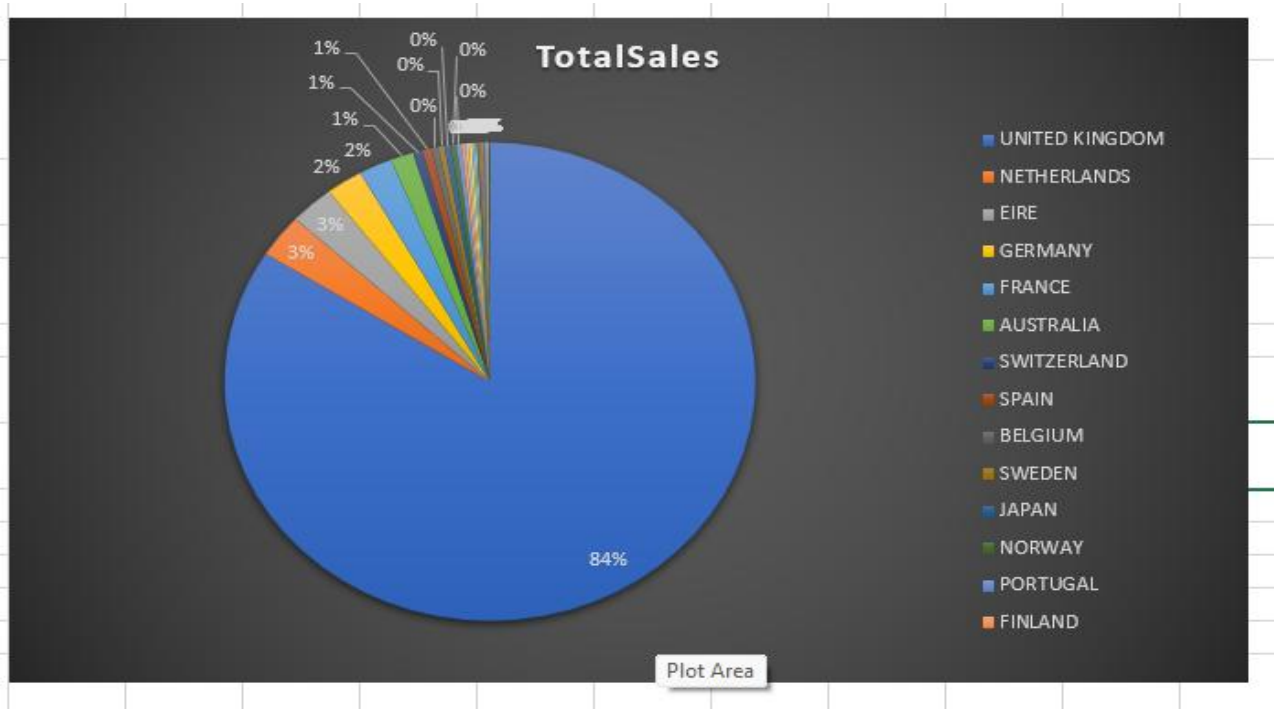
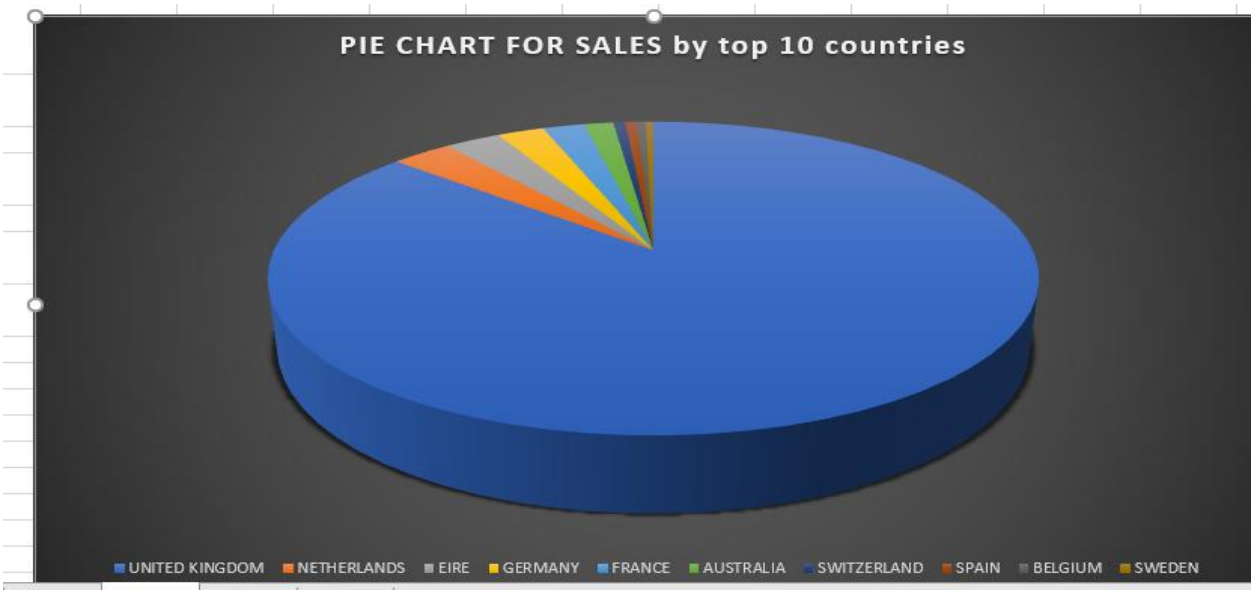


Figure 2: Distribution of Quantity Sold (Histogram)

Explanation:
The histogram shows that most transactions involve purchasing a small number of items, while only a few transactions involve very large quantities.

Figure 3: Sales Contribution by Top Countries (Pie Chart)

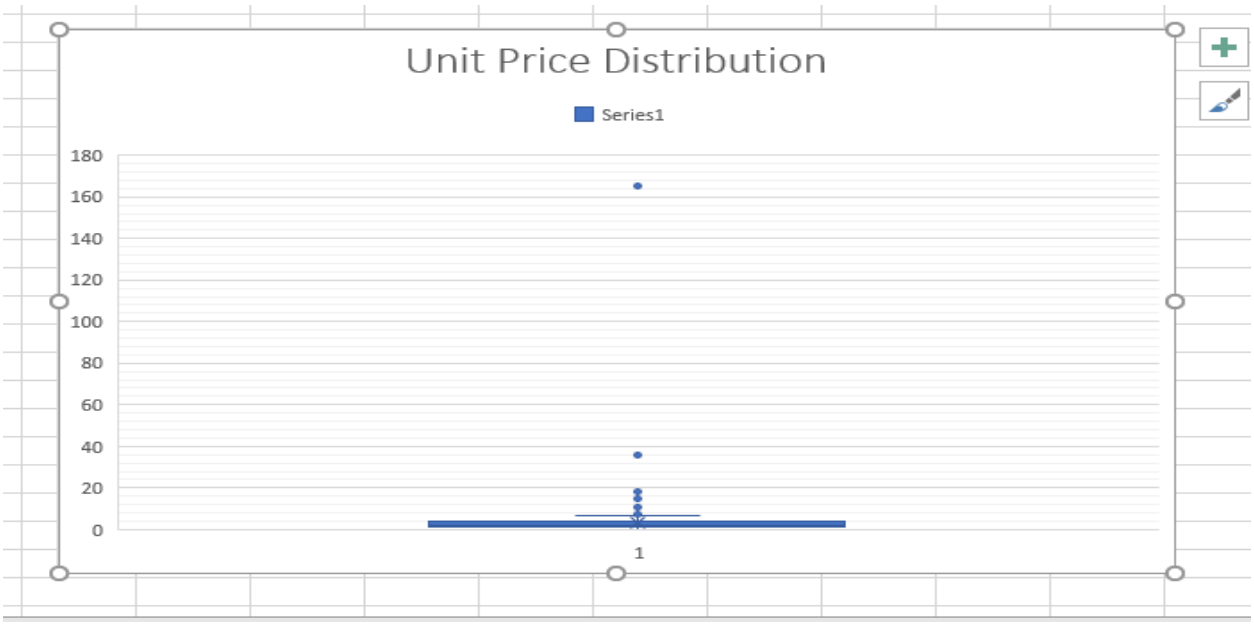


Explanation:
The pie chart illustrates the percentage contribution of the top-performing countries. Revenue is concentrated in a small number of countries.

Figure 4: Monthly Sales Trend (Line Chart)

Explanation:
The line chart reveals monthly fluctuations in sales. Peak periods may indicate seasonal demand or promotional campaigns.

Figure 5: Distribution of Unit Prices (Box Plot)



Explanation:
The box plot shows the spread of product prices and identifies outliers. Most products are sold within a narrow price range, while a few premium-priced products stand out.

These five visualizations satisfy the assignment requirement of including a **title**, **axis labels**, and a **brief explanation of the findings** for the cleaned retail sales dataset.

Task 5: Key Insights

Insight 1: The United Kingdom Generates the Highest Sales Revenue

Evidence:
The **Bar Chart: Total Sales by Country** shows that the United Kingdom contributes the largest proportion of total sales compared to all other countries.

Explanation:
This indicates that the UK is the company's primary market and contributes significantly to overall revenue. The business should continue investing in customer retention and marketing strategies within this market.

Reference:
Figure 1 – Total Sales by Country (Bar Chart)

Insight 2: Most Transactions Involve Small Purchase Quantities

Evidence:
The **Histogram of Quantity Sold** shows that most purchases fall within the lower quantity ranges, while only a few transactions involve very large quantities.

Explanation:
This suggests that the majority of customers purchase relatively small numbers of products per transaction. Large-volume purchases are uncommon and may represent wholesale or business customers.

Reference:
Figure 2 – Distribution of Quantity Sold (Histogram)

Insight 3: Sales Show Seasonal Variations

Evidence:
The **Monthly Sales Trend Line Chart** indicates that sales fluctuate throughout the year, with certain months experiencing noticeably higher revenue.

Explanation:
The variation suggests seasonal demand, promotional campaigns, or holiday shopping periods. The business can use this information for inventory planning and targeted marketing.

Reference:
Figure 4 – Monthly Sales Trend (Line Chart)

Insight 4: A Few Countries Contribute Most of the Revenue

Evidence:
The **Pie Chart of Sales by Top Countries** shows that a small number of countries account for the majority of total sales.

Explanation:
The company depends heavily on a few international markets. Expanding into underperforming regions could reduce market concentration risk.

Reference:
Figure 3 – Sales Contribution by Top Countries (Pie Chart)

Insight 5: Product Prices Contain Several Outliers

Evidence:
The **Box Plot of Unit Prices** identifies several unusually high-priced products compared to the majority of items.

Explanation:
Most products are sold within a relatively small price range, while a few premium products have much higher prices. These outliers should be investigated to determine whether they represent luxury items, pricing errors, or special products.

Reference:
Figure 5 – Unit Price Distribution (Box Plot)

Summary Report

Online Retail Dataset Analysis Report

1. Cleaning Challenges Encountered

- Removed duplicate records to improve data quality.
- Handled missing values in important fields such as `CustomerID` and `Description`.
- Converted `InvoiceDate` to the correct Date/Time format.
- Removed cancelled transactions and invalid records containing negative quantities or negative unit prices.
- Verified data types for all variables before analysis.

2. Key Exploratory Data Analysis (EDA) Findings

- The dataset contains transactions from multiple countries.
- The United Kingdom records the highest sales revenue.
- Most purchases involve relatively small quantities.
- Sales fluctuate across different months, indicating seasonal demand.
- Product prices are generally consistent, although several high-value outliers exist.

3. Top Insights

1. The United Kingdom is the company's largest revenue-generating market.
2. Most customers purchase only a few items per transaction.
3. Sales vary across the year, suggesting seasonal purchasing behavior.
4. A small number of countries account for most of the company's revenue.
5. Unit prices include a few significant outliers that may require further investigation.